Math Fair: Kinetic Physics Engine Report

Researcher: Student Researcher

Global Summary

Tracked object: Phone
Assumed mass: 0.180 kg
Total duration: 19.370 s
Maximum height: 0.197 m
Maximum modeled speed: 0.133 m/s
Average coefficient of restitution: 1.008 (std = 0.010)

The object reached a maximum height of 0.197 meters and was tracked for approximately 19.370 seconds.
The coefficient of restitution measures how much energy is conserved during bounces; in this trial it averaged about 1.008.
The motion was automatically split into piecewise segments at turning points in height, and each segment was modeled by the best-fitting function among constant, linear, and quadratic.

Segment Details

Segment 1: Quadratic Motion

```
Time range: 2.941 s -> 5.287 s
x(t) = -0.068 t + 0.391
y(t) = 0.017 t^2 + -0.005 t + 0.093
R2_x = 0.8474,  R2_y = 0.9211
```

Segment 2: Quadratic Motion

```
Time range: 5.349 s -> 6.915 s
x(t) = -0.022 t + 0.319
y(t) = -0.039 t^2 + -0.008 t + 0.182
R2_x = 0.0273,  R2_y = 0.9828
```

Segment 3: Quadratic Motion

```
Time range: 6.979 s -> 10.637 s
x(t) = 0.015 t + 0.196
y(t) = -0.014 t^2 + 0.070 t + 0.099
R2_x = 0.5050,  R2_y = 0.7434
```

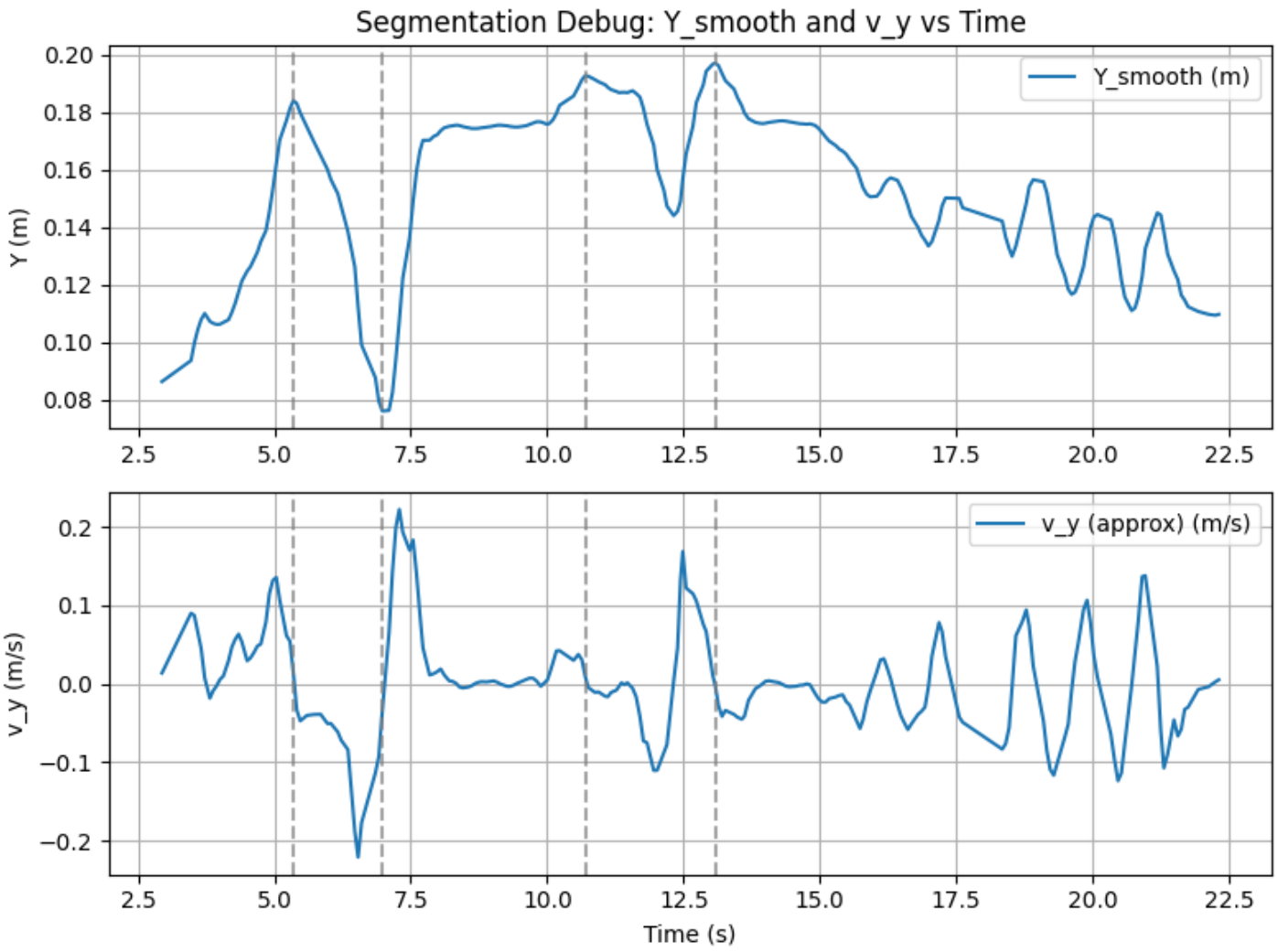
Segment 4: Quadratic Motion

```
Time range: 10.696 s -> 13.035 s
x(t) = 0.069 t + 0.216
y(t) = 0.020 t^2 + -0.056 t + 0.207
R2_x = 0.6686,  R2_y = 0.4094
```

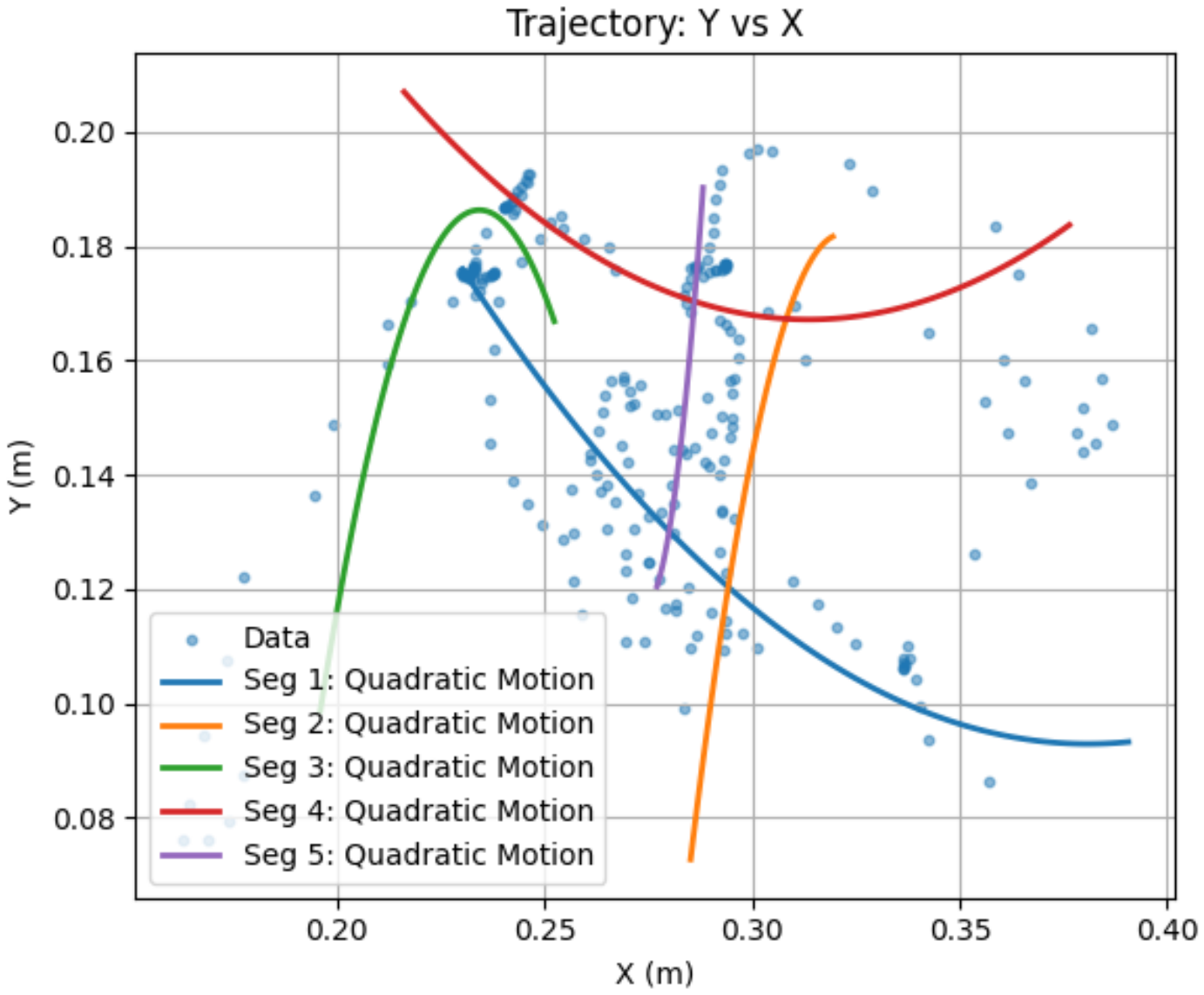
Segment 5: Quadratic Motion

```
Time range: 13.081 s -> 22.310 s
x(t) = -0.001 t + 0.288
y(t) = 0.001 t^2 + -0.012 t + 0.190
R2_x = 0.0800,  R2_y = 0.7840
```

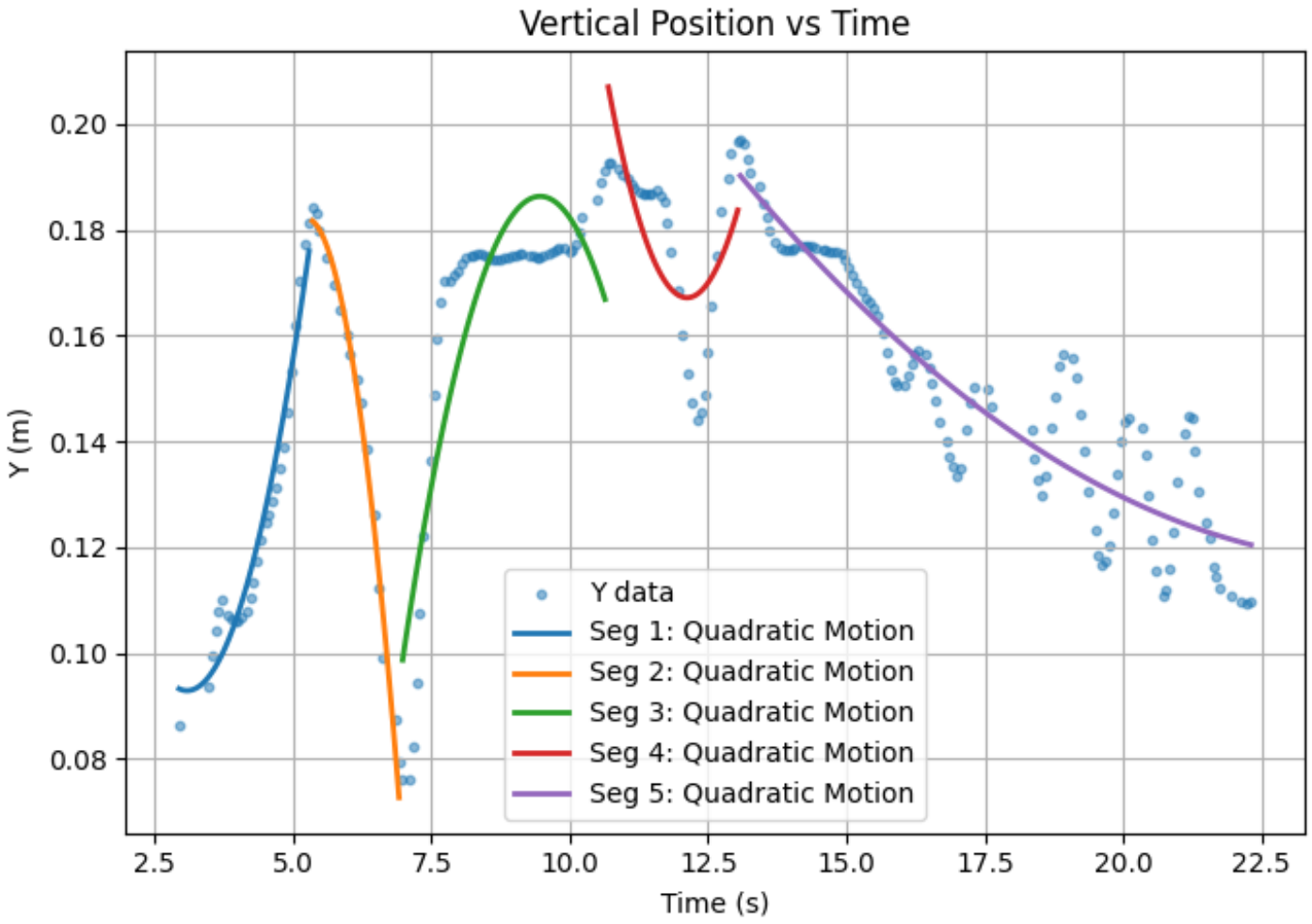
Segmentation Debug (Y and v_y vs Time)



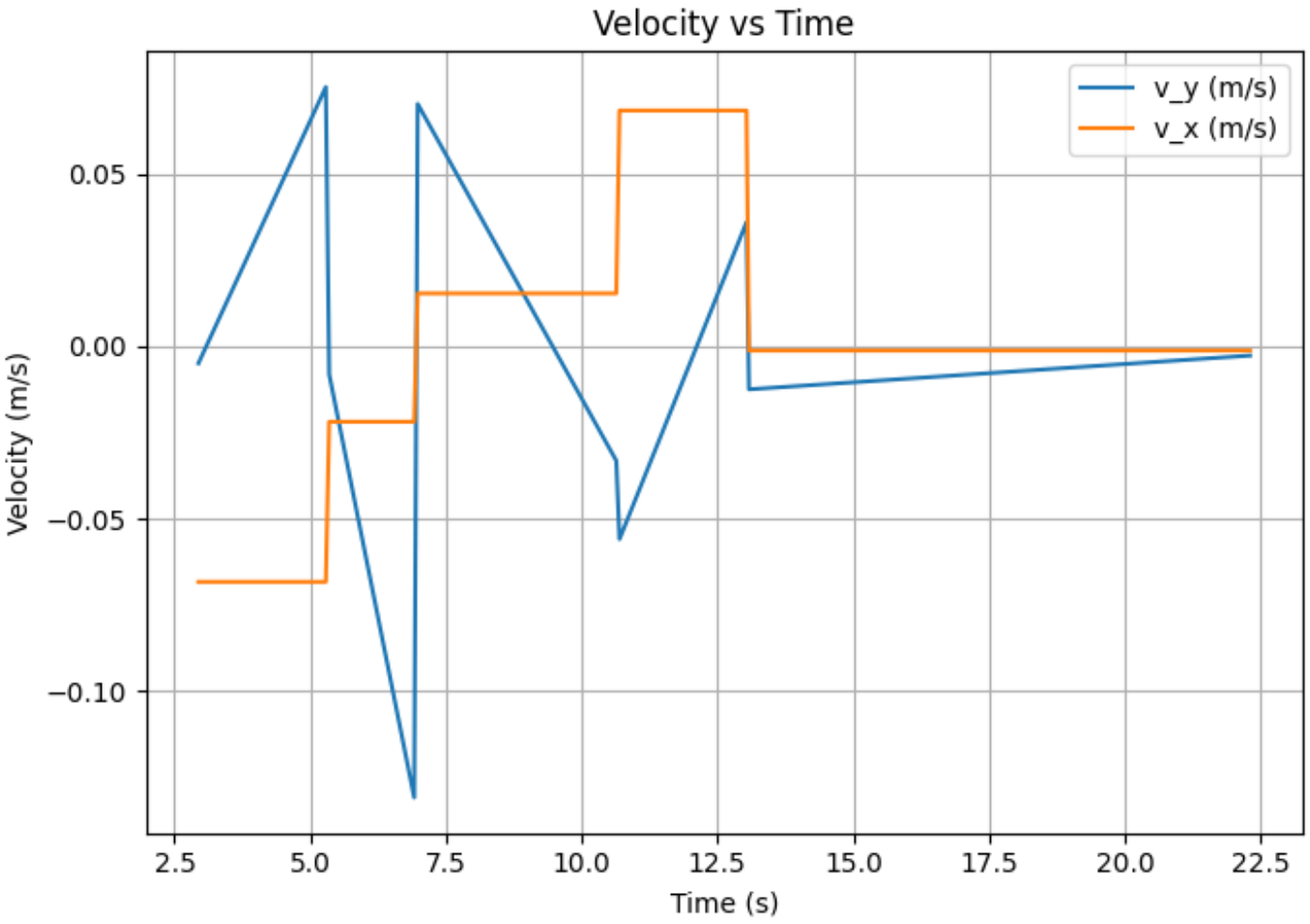
Trajectory (Y vs X)



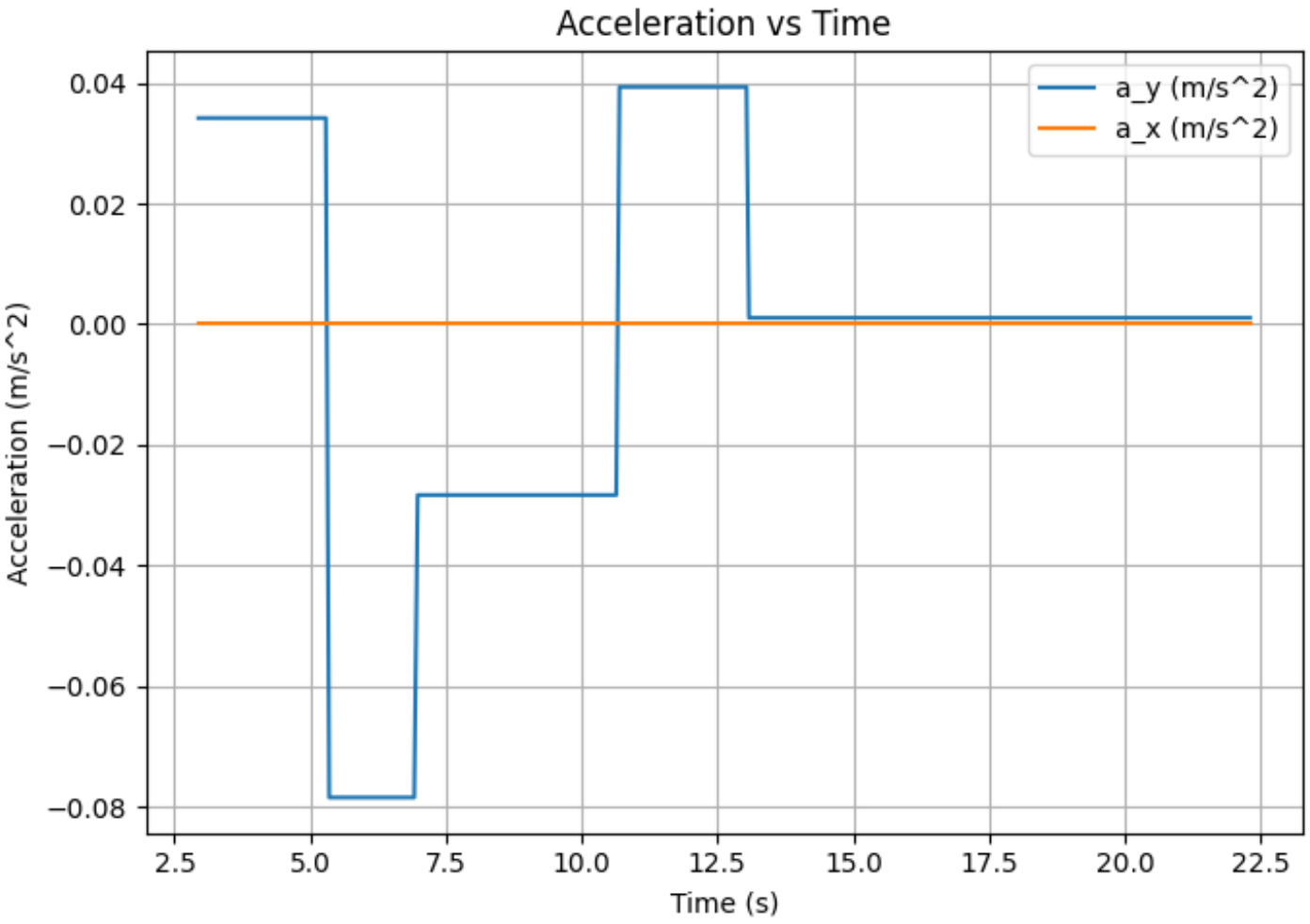
Vertical Position vs Time



Velocity vs Time



Acceleration vs Time



Energy vs Time

